

A high-resolution photograph taken from space, showing the Earth's horizon as a bright blue curve against the black void of space. A portion of a space station or satellite is visible in the foreground, featuring a white cylindrical module with a circular hatch and a long, slender boom extending upwards. The Earth's surface below is covered in a dense layer of white clouds, with some darker landmasses visible through the cloud cover.

Exam Questions

Space II

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Questions

According to Kepler's First Law, what is the shape of a planetary orbit?

- a) Circular
- b) Elliptical
- c) Parabolic
- d) Hyperbolic

Newton's Law of Universal Gravitation states that the gravitational force between two objects is:

- a) Directly proportional to their masses and inversely proportional to the square of the distance between them
- b) Directly proportional to the square of their masses and inversely proportional to their distance
- c) Independent of the distance between the objects
- d) Only dependent on the larger object's mass

Questions

Which of the following best describes Kepler's Second Law?

- a) The ratio of the squares of the orbital periods of planets is equal to the ratio of the cubes of their semi-major axes
- b) The orbits of planets are elliptical, with the Sun at one focus
- c) A line connecting a planet to the Sun sweeps out equal areas in equal time intervals
- d) Planets further from the Sun move faster than those closer

What does Kepler's Third Law allow us to determine?

- a) The speed of a planet at any given moment
- b) The relationship between the period of orbit and the semi-major axis of the orbit
- c) The gravitational force acting on the planet
- d) The eccentricity of the orbit

Questions

Two satellites orbiting Earth have masses of 500 kg and 1,000 kg respectively. If the distance between them is 200 km, what happens to the gravitational force if the distance doubles?

- a) It doubles
- b) It quadruples
- c) It halves
- d) It becomes one-fourth

Questions

The Moon orbits Earth at an average distance of 384,400 km with an orbital period of 27.3 days. If another moon were discovered orbiting Earth with twice this distance, what would be its approximate orbital period?

- a) 38.6 days
- b) 54.6 days
- c) 77.1 days
- d) 96.9 days

Questions

A satellite orbits Earth with a semi-major axis of 10,000 km and an eccentricity of 0.2. What is its closest approach to Earth (perigee distance)?

- a) 8,000 km
- b) 10,000 km
- c) 12,000 km
- d) 14,000 km

Questions

Two planets, A and B, orbit a star. Planet A has an orbital period of 2 years, while Planet B's semi-major axis is twice that of Planet A. What is the orbital period of Planet B?

- a) 4 years
- b) 5.7 years
- c) 8 years
- d) 11.3 years

Questions

A planet orbits a star in an elliptical orbit with an eccentricity of 0.5 and a semi-major axis of 150 million km. What is its farthest distance from the star (apogee)?

- a) 100 million km
- b) 150 million km
- c) 200 million km
- d) 225 million km

Questions

If the mass of the Sun were doubled while keeping Earth's distance the same, how would Earth's orbital period change?

- a) It would increase
- b) It would decrease
- c) It would remain the same
- d) It would become unpredictable

Questions

A planet is observed to have an elliptical orbit with an eccentricity of 0.8 and a semi-major axis of 500 million km. What is the planet's closest distance to its star?

- a) 100 million km
- b) 200 million km
- c) 300 million km
- d) 400 million km

Questions

Two satellites, A and B, orbit the Earth. Satellite A orbits at 10,000 km from the Earth's center, while Satellite B orbits at 40,000 km. How does Satellite B's period compare to Satellite A's?

- a) 2 times longer
- b) 4 times longer
- c) 8 times longer
- d) 16 times longer

Questions

A spacecraft in orbit around Earth suddenly increases its velocity at perigee. How does this affect its orbit?

- a) The orbit becomes more circular
- b) The orbit becomes more elliptical with a higher apogee
- c) The spacecraft moves to a lower orbit
- d) The eccentricity of the orbit decreases

Questions

A satellite is in a circular orbit around Earth with a radius of 10,000 km. If it moves to an elliptical orbit with an apogee of 20,000 km, how does its velocity at perigee compare to its original velocity?

- a) Increases
- b) Decreases
- c) Remains the same
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Answers

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- a) Circular
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